8A 1 ELECTRON LOSS AND GAIN

SPECIFICATION REFERENCE: 8.5 8.6 8.7

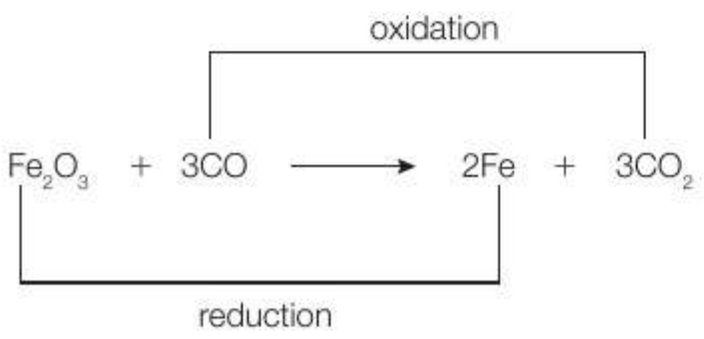
LEARNING OBJECTIVES

- Understand oxidation and reduction in terms of electron transfer and changes in oxidation number.
- Understand the application of these ideas to reactions of s-block and p-block elements.
- Know that oxidising agents gain electrons and that reducing agents lose electrons.
- Understand that a disproportionation reaction involves an element in a single species being simultaneously oxidised and reduced.

BACKGROUND TO OXIDATION AND REDUCTION

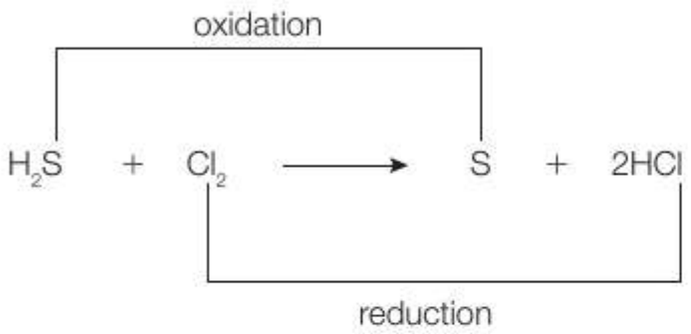
You originally knew 'oxidation' as the addition of oxygen and 'reduction' as the removal of oxygen.

The reaction between iron(III) oxide and carbon monoxide that takes place in the blast furnace provides an example of each (**fig A**).



▲ **fig A** Example of oxidation and reduction during the reaction between iron(III) oxide and carbon monoxide.

We then expanded these definitions so that oxidation is described as the removal of hydrogen and reduction is described as the addition of hydrogen. The reaction between chlorine and hydrogen sulfide provides an example (**fig B**).



▲ **fig B** Example of oxidation and reduction during the reaction between chlorine and hydrogen sulfide.

A reaction that involves both reduction and oxidation is called a **redox reaction**.

OXIDATION AND REDUCTION IN TERMS OF ELECTRON LOSS AND GAIN

ELECTRON TRANSFER IN REDOX REACTIONS

The approach preferred by chemists today to describe oxidation and reduction involves the transfer of electrons.

When magnesium burns in oxygen it forms magnesium oxide:

$$2Mg + O_2 \rightarrow 2MgO$$

The magnesium has been oxidised because it has gained oxygen. The oxygen must have been reduced, but by what definition?

The reaction results in the formation of Mg^{2+} and O^{2-} ions. Each of the two magnesium atoms has lost two electrons:

$$Mg \rightarrow Mg^{2+} + 2e^-$$

The oxygen molecule has gained four electrons to become oxide ions:

$$O_2 + 4e^- \rightarrow 2O^{2-}$$

The magnesium has been *oxidised* because it has lost electrons.

Oxygen has been *reduced* because it has gained electrons.

DEFINITIONS OF OXIDATION AND REDUCTION

We now have new definitions of oxidation and reduction:

Oxidation is the *loss* of electrons.

Reduction is the *gain* of electrons.

These new definitions are easily remembered using the mnemonic **OIL RIG**.

Oxidation

Is

Loss (of electrons)

Reduction

Is

Gain (of electrons)

We can now describe the reduction of iron(III) oxide in the blast furnace as the gain of three electrons by the iron(III) ion in the oxide:

$$Fe^{3+} + 3e^- \rightarrow Fe$$

The carbon monoxide must have been oxidised (it gains oxygen), but it is not easy to see how it has lost electrons because carbon monoxide and carbon dioxide are both covalent compounds.

OXIDISING AND REDUCING AGENTS

In the reaction between magnesium and oxygen to form magnesium oxide, the oxygen has oxidised the magnesium.

Therefore, oxygen is an **oxidising agent**.

The magnesium has reduced the oxygen and is therefore a **reducing agent**.

EXAM HINT

An oxidising agent is reduced.

DISPROPORTIONATION

Consider the following reaction, which occurs if copper(I) oxide is added to dilute sulfuric acid:

$$Cu_2O(s) + H_2SO_4 \rightarrow CuSO_4(aq) + Cu(s) + H_2O(l)$$

One of the Cu^+ ions in copper(I) oxide has lost an electron to become Cu^{2+}. At the same time, the other Cu^+ ion has gained an electron to become Cu. Both oxidation and reduction have occurred but the *same element*, copper, is involved in both changes.

This is an example of a **disproportionation reaction**.

CHECKPOINT

1. In the reaction:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

(a) Which species has been oxidised and which species has been reduced?

(b) Write ionic half-equations to represent both the oxidation and the reduction reactions.

2. In each of the following reactions identify whether the underlined species has been oxidised, reduced or neither.

(a) $\underline{Al}(s) + 1\frac{1}{2}Cl_2(g) \rightarrow AlCl_3(s)$

(b) $4\underline{Na}(s) + TiO_2(s) \rightarrow 2Na_2O(s) + Ti(s)$

(c) $\underline{Ag^+}(aq) + Cl^-(aq) \rightarrow AgCl(s)$

(d) $Cl_2(aq) + 2\underline{Br^-}(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$

(e) $\underline{Cu}O(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2O(l)$

(f) $\underline{Mn}O_2(s) + 4HCl(aq) \rightarrow MnCl_2(aq) + Cl_2(aq) + 2H_2O(l)$

3. In each of the following reactions, identify the species that has been oxidised. In each case, justify your answer.

(a) $Fe(s) + H_2SO_4(aq) \rightarrow FeSO_4(aq) + H_2(g)$

(b) $Na(s) + \frac{1}{2}H_2(g) \rightarrow NaH(s)$

(c) $CuO(s) + Cu(s) \rightarrow Cu_2O(s)$

(d) $2Fe(OH)_2(s) + \frac{1}{2}O_2(aq) + H_2O(l) \rightarrow 2Fe(OH)_3(s)$

(e) $2V^{3+}(aq) + Zn(s) \rightarrow 2V^{2+}(aq) + Zn^{2+}(aq)$

SUBJECT VOCABULARY

redox reaction a reaction that involves both reduction and oxidation

oxidation when the oxidation number of an element increases; the loss of electrons

reduction when the oxidation number of an element decreases; the gain of electrons

oxidising agent a species (atom, molecule or ion) that oxidises another species by removing one or more electrons; when an oxidising agent reacts it gains electrons and is, therefore, reduced

reducing agent a species that reduces another species by adding one or more electrons; when a reducing agent reacts it loses electrons and is, therefore, oxidised

disproportionation reaction a reaction involving the simultaneous oxidation and reduction of an element in a single species

8A 2 ASSIGNING OXIDATION NUMBERS

SPECIFICATION REFERENCE: 8.1 8.2 8.9

LEARNING OBJECTIVES

- Know what is meant by the term oxidation number and understand the rules for assigning oxidation numbers.
- Be able to calculate the oxidation number of elements in compounds and ions, including in peroxides and metal hydrides.
- Understand that metals, in general, form positive ions by loss of electrons with an increase in oxidation number whereas non-metals, in general, form negative ions by gain of electrons with a decrease in oxidation number.

OXIDATION NUMBER

So far we have restricted our discussion of oxidation and reduction to atoms and ions. For redox reactions involving these species it is easy to see which species are losing and which are gaining electrons. However, many compounds are covalent and for them a simple treatment involving ions is not appropriate.

To get around this difficulty, the concept of **oxidation number** has been developed.

For example, in MgO the oxidation number of magnesium is +2, since the charge on the magnesium ion is 2^+. Similarly, the oxidation number of oxygen is −2.

In SO_2, the oxidation number of the sulfur is +4 because if the compound were fully ionic the sulfur ion would have a charge of 4^+. The oxidation number of oxygen is once again −2.

RULES FOR DETERMINING THE OXIDATION NUMBER

Here are some rules to help you calculate the oxidation number.

- The oxidation number of an uncombined element is zero.
- The sum of the oxidation numbers of all the elements in a neutral compound is zero.
- The sum of the oxidation numbers of all the elements in an ion is equal to the charge on the ion.
- The more electronegative element in a substance is given a negative oxidation number.
- The oxidation number of fluorine is always −1.
- The oxidation number of hydrogen is +1, except when combined with a less electronegative element. Then it becomes −1.
- The oxidation number of oxygen is −2, except in peroxides where it is −1 and when combined with fluorine when it is +2.

The best way to get used to these rules is to put them into practice, which is what we shall now do.

WORKED EXAMPLE 1

Deduce the oxidation number of chlorine in:

(a) NaCl

(b) NaClO

(c) $NaClO_3$.

Answers

(a) The oxidation number of Na is +1. The two oxidation numbers must add up to zero, so the oxidation number of Cl must be −1.

(b) The oxidation number of Na is +1 and the oxidation number of O is −2. The oxidation numbers must add up to zero, so the oxidation number of Cl must be +1.

(c) The oxidation number of Na is +1 and the oxidation number of O is −2. The oxidation numbers must add up to zero, so the oxidation number of Cl must be +5.

$$+1 + (3 \times -2) + x = 0$$

$$x = 0 - 1 - (-6) = +5$$

WORKED EXAMPLE 2

Deduce the oxidation number of nitrogen in:

(a) NH_3

(b) NO_2^-

(c) NO_3^-.

Answers

(a) N is more electronegative than H and must therefore have a negative oxidation number. The oxidation number of H is therefore +1. The oxidation numbers must add up to zero, so the oxidation number of N is −3.

$x + (3 \times 1) = 0$

$x = -3$

(b) O is more electronegative than N, so the oxidation number of O is −2. The oxidation numbers must add up to −1 (the charge on the ion), so the oxidation number of N is +3.

$x + (2 \times -2) = -1$

$x = +3$

(c) O is more electronegative than N, so the oxidation number of O is −2. The oxidation numbers must add up to −1 (the charge on the ion), so the oxidation number of N is +5.

$x + (3 \times -2) = -1$

$x = +5$

CHECKPOINT

1. Calculate the oxidation number of the underlined element in each of the following species.

$\underline{S}O_2$	$\underline{S}O_3$	$H_2\underline{S}$	$\underline{S}O_3^{2-}$	$\underline{S}O_4^{2-}$	$\underline{S}_2O_3^{2-}$
$\underline{Cr}O_4^{2-}$	$\underline{Cr}_2O_7^{2-}$	$\underline{Mn}O_4^{2-}$	$\underline{Mn}O_4^-$	$\underline{V}O^{2+}$	$\underline{V}O_2^+$
$\underline{Cl}O^-$	$\underline{Cl}O_2^-$	$\underline{Cl}O_3^-$	$\underline{Cl}O_4^-$	$[\underline{Cu}Cl_4]^{2-}$	$[\underline{V}(H_2O)_6]^{2+}$
$H_2\underline{O}_2$	$\underline{O}F_2$	$Na\underline{H}$	$Ba\underline{O}_2$		

SUBJECT VOCABULARY

oxidation number the charge that an ion has, or the charge that it would have if the species were fully ionic

8A 3 RECOGNISING REACTIONS USING OXIDATION NUMBERS

SPECIFICATION REFERENCE: 8.5 8.8

LEARNING OBJECTIVES

- Understand oxidation and reduction in terms of electron transfer and changes in oxidation number, and the application of these ideas to reactions of s-block and p-block elements.
- Know that the oxidation number is a useful concept in terms of the classification of reactions as redox and as disproportionation.

USING OXIDATION NUMBERS TO CLASSIFY REACTIONS

Consider the following ionic half-equations:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

$2I^-(aq) \rightarrow I_2(aq) + 2e^-$

In all three cases, electrons have been lost so the reactions are oxidations. But also notice that the oxidation number of the element has increased in each case.

- Zn has increased from 0 to +2.
- Fe has increased from +2 to +3.
- I has increased from −1 to 0.

This leads to another definition of **oxidation** and **reduction**.

Consider the following ionic half-equations:

$Cl_2(aq) + 2e^- \rightarrow 2Cl^-(aq)$	Cl changes from 0 to −1
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$	Mn changes from +7 to +2
$2H_2O(l) + 2e^- \rightarrow 2OH^-(aq) + H_2(g)$	H changes from +1 to 0

In each case, the oxidation number of one of the elements involved has decreased, so reduction has taken place.

This concept can now be applied to full equations.

EXAM HINT

Notice that the higher oxidation numbers for metal ions in solution are always in the form of oxyions, e.g., in CrO_4^{2-} Cr has an oxidation number of +6.

WORKED EXAMPLE

Use oxidation numbers to show whether the following reactions are examples of redox reactions.

(a) $H_2S(g) + Cl_2(g) \rightarrow 2HCl(g) + S(s)$

Oxidation numbers: +1 −2 0 +1 −1 0

Answer

The oxidation number of S has increased (−2 to 0), so it has been oxidised.

The oxidation number of Cl has decreased (0 to −1), so it has been reduced.

Since both oxidation and reduction have taken place, the reaction is classified as a redox reaction.

(b) $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

Oxidation numbers: +1 −2 +1 +1 −1 +1 −1 +1 −2

Answer

There is no change in oxidation number for any of the elements involved in the reaction, so this is not an example of a redox reaction.

(c) $2NaOH(aq) + Cl_2(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$

Oxidation numbers: +1 −2 +1 0 +1 −1 +1 +1 −2 +1 −2

Answer

The oxidation number of Cl has both increased (0 in Cl_2 to +1 in NaClO) and decreased (0 in Cl_2 to −1 in NaCl), so this is an example of a redox reaction.

It is also an example of a **disproportionation reaction**.

REACTIONS OF S-BLOCK ELEMENTS

We can make some predictions about the reactions of s-block elements in terms of redox.

Reactive metals in Groups 1 and 2 lose electrons when they react. For example, the Group 1 element sodium loses one electron, which can be represented in this ionic half-equation:

$$Na \rightarrow Na^+ + e^-$$

At the same time, you can also recognise that this involves an increase in oxidation number from 0 to +1.

Magnesium, a Group 2 metal, reacts in a similar way:

$$Mg \rightarrow Mg^{2+} + 2e^-$$

This involves an increase in oxidation number from 0 to +2.

You will learn more about these reactions in **Topic 8B**.

REACTIONS OF P-BLOCK ELEMENTS

We can also make some predictions about the reactions of p-block elements in terms of redox. Unfortunately, it is not as straightforward as with s-block elements. This is because the p-block contains elements with different characteristics. For example, the Group 3 element aluminium is a metal with a reactivity nearly as high as magnesium, so it reacts in a similar way:

$$Al \rightarrow Al^{3+} + 3e^-$$

This involves an increase in oxidation number from 0 to +3.

Now consider the Group 7 element fluorine, which is the most reactive halogen. The half-equation for its reaction is:

$$F_2 + 2e^- \rightarrow 2F^-$$

This involves a decrease in oxidation number from 0 to –1.

Nitrogen is a p-block element in Group 5, with an electronegativity higher than aluminium, but lower than fluorine. What predictions can we make about the change in oxidation number in its reactions? This depends on which element it reacts with.

For example, nitrogen reacts with the Group 1 metal sodium as follows:

$$6Na + N_2 \rightarrow 2Na_3N$$

In this redox reaction, the oxidation number of sodium increases from 0 to +1, and that of nitrogen decreases from 0 to –3.

Now consider nitrogen's reaction with fluorine:

$$N_2 + 3F_2 \rightarrow 2NF_3$$

As always, the oxidation number of fluorine decreases from 0 to –1, but this time the oxidation number of nitrogen increases from 0 to +3. You will learn more about these reactions in **Topic 8C**.

CHECKPOINT

1. Copy the following table. Use oxidation numbers to complete it. The first example has been done for you.

EQUATION	REDOX REACTION (✓ OR ✗)	DISPROPORTIONATION (✓ OR ✗)	ELEMENT OXIDISED	ELEMENT REDUCED
$Mg + 2HCl \rightarrow MgCl_2 + H_2$ 0 +1 –1 +2 –1 0	✔	✘	Mg	H
$CaO + H_2O \rightarrow Ca(OH)_2$				
$2H_2O_2 \rightarrow 2H_2O + O_2$				
$KOH + HNO_3 \rightarrow KNO_3 + H_2O$				
$Cl_2 + H_2O \rightarrow HCl + HClO$				

SUBJECT VOCABULARY

oxidation when the oxidation number of an element *increases*; the loss of electrons

reduction when the oxidation number of an element *decreases*; the gain of electrons

disproportionation reaction a reaction involving the simultaneous oxidation and reduction of an element in a single species

8A 4 OXIDATION NUMBERS AND NOMENCLATURE

SPECIFICATION REFERENCE: 8.3 8.4

LEARNING OBJECTIVES

- Be able to indicate the oxidation number of an element in a compound or ion, using a Roman numeral.
- Be able to write formulae given oxidation numbers.

SYSTEMATIC NAMES

When an element can have more than one oxidation state, the names of its compounds and its ions often include the oxidation number of the element, written as a Roman numeral in brackets. This name is often referred to as the 'systematic name'.

Table A shows some examples.

FORMULA OF COMPOUND OR ION	RELEVANT OXIDATION NUMBER	SYSTEMATIC NAME OF COMPOUND OR ION
$FeCl_2$	Fe +2	iron(II) chloride
$FeCl_3$	Fe +3	iron(III) chloride
$KMnO_4$	Mn +7	potassium manganate(VII)
K_2MnO_4	Mn +6	potassium manganate(VI)
CrO_4^{2-}	Cr +6	chromate(VI) ion
$Cr_2O_7^{2-}$	Cr +6	chromate(VI) ion

table A Examples of how to indicate the oxidation number of an element in a compound or ion using the systematic name.

WHEN TO USE SYSTEMATIC NAMES

We often use systematic names in chemistry so that we can be specific about the compounds and ions we are referring to. However, in the wider world the numbers are often left out.

For example:

- the systematic name for Na_2SO_4 is sodium sulfate(VI), but it is often just called sodium sulfate
- Na_2SO_3 should be labelled sodium sulfate(IV), but the name sodium sulfite is still commonly used
- SO_2 and SO_3 are more commonly referred to as sulfur dioxide and sulfur trioxide, rather than sulfur(IV) oxide and sulfur(VI) oxide, respectively.

You should use systematic names as often as possible, particularly during your studies.

WRITING FORMULAE WHEN YOU HAVE THE OXIDATION NUMBER

The other skill you need to develop is to work backwards from the oxidation number to deduce the formula of the compound or ion concerned.

WORKED EXAMPLE 1

Deduce the formula for iron(II) sulfate.

Answer

The formula of the iron(II) ion is Fe^{2+}.

The formula of the sulfate ion is SO_4^{2-}.

So the two ions are present in a 1 : 1 ratio to produce a neutral compound, giving the formula $FeSO_4$.

WORKED EXAMPLE 2

Deduce the formula for iron(III) sulfate.

Answer

The formula of the iron(III) ion is Fe^{3+}.

The formula of the sulfate ion is SO_4^{2-}.

So the two ions are present in a 2 : 3 ratio to give a neutral compound, giving the formula $Fe_2(SO_4)_3$.

CHECKPOINT

1. Give the systematic name for each of the following compounds:

(a) PCl_3

(b) PCl_5

(c) V_2O_5

(d) NaClO

(e) $NaClO_3$

2. Give the systematic name for each of the following ions:

(a) NO_3^-

(b) NO_2^-

(c) ClO_4^-

(d) VO_2^+

(e) VO_2^+

3. Deduce the formula for each of the following compounds:

(a) copper(I) oxide

(b) copper(II) oxide

(c) chromium(III) sulfate(VI)

(d) lead(IV) iodide

(e) cobalt(III) nitrate(V)

4. Why is it not necessary to refer to sodium chloride as sodium(I) chloride, or magnesium oxide as magnesium(II) oxide?

8A 5 CONSTRUCTING FULL IONIC EQUATIONS

SPECIFICATION REFERENCE 8.10

LEARNING OBJECTIVES

- Be able to write ionic half-equations and use them to construct full ionic equations.

BALANCING BY COUNTING ELECTRONS

STRAIGHTFORWARD EXAMPLES

When solid zinc is added to an aqueous solution of copper(II) sulfate, the following two changes take place:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ and $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Both ionic half-equations involve two electrons, so to construct the full ionic equation for this reaction you simply add together the two half-equations so that the electrons cancel out:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$
$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

Now let's try an example where the electrons are not the same in the two ionic half-equations.

When chlorine gas is bubbled into an aqueous solution of iron(II) chloride, the iron(II) ions are oxidised to iron(III) ions and the chlorine molecules are reduced to chloride ions.

The two ionic half-equations are:

$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$ and $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$

This time, one of the half-equations contains one electron, while the other contains two electrons. Before these can be added together to produce a full equation, the equation containing Fe^{2+} must be multiplied by 2.

$$2Fe^{2+}(aq) \rightarrow 2Fe^{3+}(aq) + 2e^-$$
$$Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$$

$$2Fe^{2+}(aq) + Cl_2(g) \rightarrow 2Fe^{3+}(aq) + 2Cl^-(aq)$$

MORE COMPLICATED EXAMPLES

Here is something a little more difficult.

When an acidified aqueous solution of potassium manganate(VII) is added to an aqueous solution of iron(II) sulfate, the following two changes occur:

$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$ and
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$

In order to balance the electrons, the first half-equation must be multiplied by 5:

$$5Fe^{2+}(aq) \rightarrow 5Fe^{3+}(aq) + 5e^-$$
$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$$

$$5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$$

EXAM HINT

Acidified potassium manganate(VII) is very commonly used as an oxidising agent in analytical chemistry. It will be useful to learn this half equation by heart.

Here's another challenging example.

The reaction between aqueous acidified potassium manganate(VII) and hydrogen peroxide involves two changes represented by the following ionic half-equations:

$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$ and
$H_2O_2(aq) \rightarrow 2H^+(aq) + O_2(g) + 2e^-$

The lowest common multiple of 2 and 5 is 10. This means that the first half-equation should be multiplied by 2 and the second by 5 before they are added together.

$$2MnO_4^-(aq) + 16H^+(aq) + 10e^- \rightarrow 2Mn^{2+}(aq) + 8H_2O(l)$$
$$5H_2O_2(aq) \rightarrow 10H^+(aq) + 5O_2(g) + 10e^-$$

$$2MnO_4^-(aq) + 16H^+(aq) + 5H_2O_2(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 10H^+(aq) + 5O_2(g)$$

The electrons have now been cancelled out, but we are left with an equation that has H^+ ions on both sides of the equation in unequal numbers. We now have to cancel these out so that they are present on only one side. To do this, you subtract the $10H^+$ on the right-hand side from the $16H^+$ on the left-hand side to give the final equation:

$$2MnO_4^-(aq) + 6H^+(aq) + 5H_2O_2(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5O_2(g)$$

EXAM HINT

This final equation shows that this reaction takes place in acid conditions. Many redox reactions are pH sensitive.

BALANCING USING OXIDATION NUMBERS

WORKED EXAMPLE 1

Use oxidation numbers to balance the following equation:

....$SO_2(g)$ + ...$H_2O(l)$ + ...$Ag^+(aq)$ → .... $SO_4^{2-}(aq)$ + ...$H^+(aq)$ + ...$Ag(s)$

Answer

Identify the elements whose oxidation numbers have changed.

In this case:

- S changes from +4 to +6; this is a '2 electron' change
- Ag changes from +1 to 0; this is a '1 electron' change.

So the ratio of SO_2 to Ag^+ is 1 : 2. This gives:

$SO_2(g)$ + ...$H_2O(l)$ + $2Ag^+(aq)$ → $SO_4^{2-}(aq)$ + ...$H^+(aq)$ + $2Ag(s)$

We now need to balance the H and O atoms.

This gives:

$SO_2(g)$ + $2H_2O(l)$ + $2Ag^+(aq)$ → $SO_4^{2-}(aq)$ + $4H^+(aq)$ + $2Ag(s)$

Lastly, check the equation for balanced charges.

- The total charge on the left-hand side is 2+.
- The total charge on the right-hand side is also 2+ (−2 + +4).

The equation is now balanced.

WORKED EXAMPLE 2

Use oxidation numbers to balance the following equation:

....Fe^{2+} + ...ClO_3^- + ...H^+ → .... Fe^{3+} + ...Cl^- + ...H_2O

Answer

Identify the elements whose oxidation numbers have changed.

In this case:

Fe changes from +2 to +3; this is a '1 electron' change

Cl changes from +5 to −1; this is a '6 electron' change.

So the ratio of Fe to ClO_3^- is 6 : 1. This gives:

$6Fe^{2+}$ + ClO_3^- + ...H^+ → $6Fe^{3+}$ + Cl^- + ...H_2O

Once again, balance the H and O atoms.

This gives:

$6Fe^{2+}$ + ClO_3^- + $6H^+$ → $6Fe^{3+}$ + Cl^- + $3H_2O$

Check the charges.

- The total charge on the left-hand side is +17((6 × +2) + −1 + (6 × +1)).
- The total charge on the right-hand side is +17((6 × +3) + −1).

The equation is now balanced.

CHECKPOINT

1. Use each pair of ionic half-equations to construct a full ionic equation. Include state symbols.

(a) $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ and $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$

(b) $\frac{1}{2}I_2(aq) + e^- \rightarrow I^-(aq)$ and $2S_2O_3^{2-}(aq) \rightarrow S_4O_6^{2-}(aq) + 2e^-$

(c) $MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$ and $Ce^{3+}(aq) \rightarrow Ce^{4+}(aq) + e^-$

(d) $Cr_2O_7^{2-}(aq) + 14H^+(l) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$ and $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

(e) $FeO_4^{2-}(aq) + 8H^+(aq) + 3e^- \rightarrow Fe^{3+}(aq) + 4H_2O(l)$ and $C_2O_4^{2-}(aq) \rightarrow 2CO_2(g) + 2e^-$

2. Use oxidation numbers to balance each of the following equations.

(a) ....$Cu(s)$ + ...$H^+(aq)$ + ...$NO_3^-(aq)$ → ....$Cu^{2+}(aq)$ + ...$H_2O(l)$ + ...$NO(g)$

(b) ....$Cu(s)$ + ...$H^+(aq)$ + ...$NO_3^-(aq)$ → ....$Cu^{2+}(aq)$ + ...$H_2O(l)$ + ...$NO_2(g)$

(c) ....$Cl_2(g)$ + ...$OH^-(aq)$ → ....$Cl^-(aq)$ + ...$ClO_3^-(g)$ + ...$H_2O(l)$